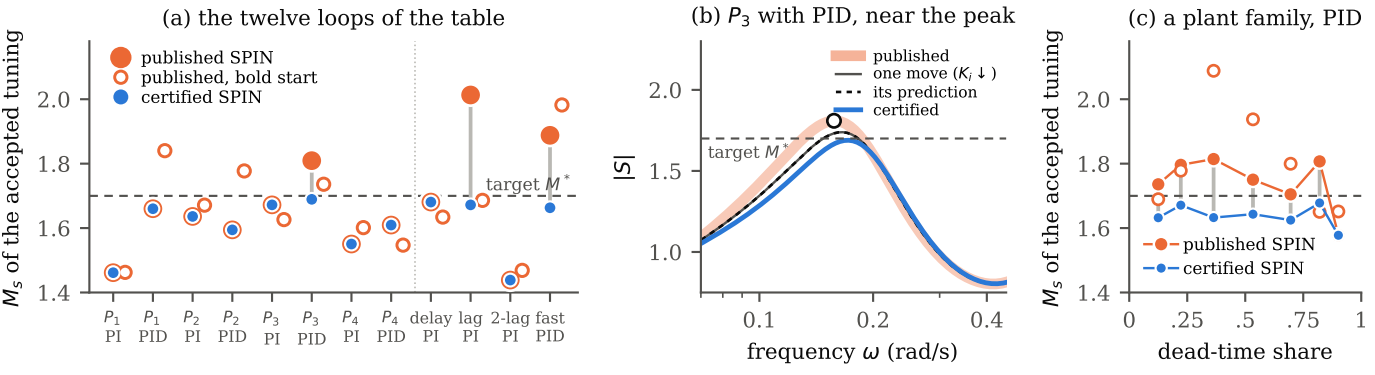
(a) the twelve loops of the table
published SPIN
published, bold start
certified SPIN
target $M^*$
$M_s$ of the accepted tuning
2.0
1.8
1.6
1.4
$P_1$ PI
$P_1$ PID
$P_2$ PI
$P_2$ PID
$P_3$ PI
$P_3$ PID
$P_4$ PI
$P_4$ PID
delay PI
lag PI
2-lag PI
fast PID
(b) $P_3$ with PID, near the peak
published
one move ($K_i \downarrow$)
its prediction
certified
target $M^*$
|S|
2.0
1.5
1.0
0.1
0.2
0.4
frequency $\omega$ (rad/s)
(c) a plant family, PID
$M_s$ of the accepted tuning
2.0
1.8
1.6
1.4
published SPIN
certified SPIN
0
.25
.5
.75
1
dead-time share